



NCCER Module 22102



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Heavy Equipment Safety





Section Three

Heavy Equipment Safety

Objective

3. Identify and describe safety practices associated with heavy equipment operation.
 - a. List job site and equipment safety requirements.
 - b. Explain the safety hazards associated with weather.
 - c. Explain safety practices associated with equipment maintenance.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

3. Demonstrate how to use various types of personal protective equipment (PPE):
 - Hard hat
 - Safety glasses
 - Ear protection
 - Gloves
 - Safety harness
 - Reflective safety vest



3.0.0 Heavy Equipment Safety

- Never use a piece of equipment that you are not trained to use.
- OSHA and MSHA require employers to train every employee in the safe operation and servicing of all equipment that the employee will use.



BULLDOZER



SCRAPER



EXCAVATOR AND OFF-ROAD TRUCK



BACKHOE WITH FRONT LOADER



MOTOR GRADER



SOIL COMPACTOR

3.1.0 Job Site and Equipment Safety Requirements

- Safety on the job is everyone's responsibility.
- The equipment operator has control over equipment operation, inspection, lubrication, and maintenance.



3.1.1 Power Lines (1 of 2)

- Equipment with tall booms, such as excavators and telescoping-boom forklifts, runs the risk of coming into contact with live power lines.
- Always maintain at least the minimum distance that must be maintained from power lines based on the voltage level.



3.1.1 Power Lines (2 of 2)

If a vehicle comes into contact with a power line:

- Do not touch any metal inside the vehicle.
- Stay inside the vehicle.
- If forced to leave, jump from the vehicle with both feet together.

**If a power line is down, call emergency services.
Always assume that any power line is live.**



3.1.2 Equipment Safety

- Before using any equipment, the operator should inspect around and under it for visible evidence of potential safety hazards or operational problems.
- Unsafe conditions must either be corrected immediately or reported.



3.1.3 Personal Safety

- Always wear the appropriate PPE.
- Wear close-fitted clothing and tie back long hair.
- Know the location of first-aid and emergency equipment.



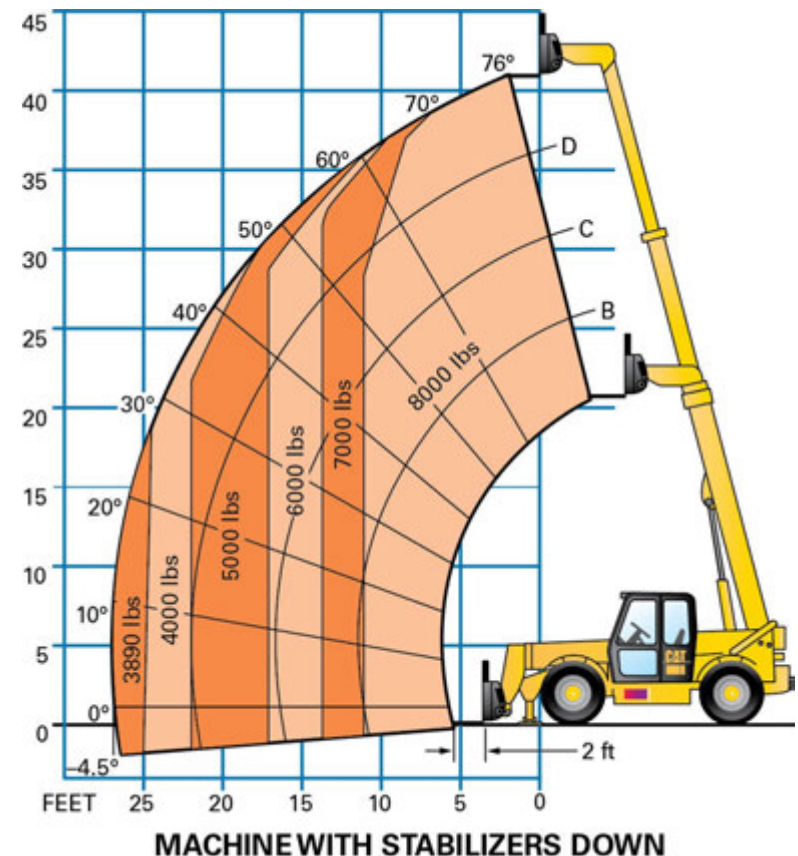
3.1.4 Moving Equipment Safely (1 of 2)

- Many injuries and deaths happen during the movement of heavy equipment.
- Each piece of heavy equipment has its own unique set of hazards and **pinch points**.



3.1.4 Moving Equipment Safely (2 of 2)

- Tipping can be caused by improper loading, unsafe driving, or unstable terrain.
- These factors affect the **center of gravity**.
- Moving the load up and down, forward and back affects the load capacity of the forklift.



3.2.0 Weather-Related Safety Hazards

Under certain environmental conditions, such as extremely hot or cold weather, work can become uncomfortable and possibly dangerous.



3.2.1 Cold Weather

- The amount of injury caused by exposure to abnormally cold temperatures depends on wind speed, length of exposure, temperature, and humidity.
- Freezing is increased by wind and humidity or a combination of the two factors.
- Wear proper clothing.
- Be careful of ice on ladders and walkway surfaces.



3.2.2 Cold Exposure Symptoms and Treatment

Symptoms include shivering, numbness, low body temperature, drowsiness, weak muscles.

To treat cold exposure:

- Notify supervision.
- Get to a warm, inside area as quickly as possible.
- Remove wet or frozen clothing.
- Rewarm by adding dry clothing or wrapping in a blanket.
- Drink warm, not hot, liquids, and do not drink alcohol.
- Check for frostbite.



3.2.3 Symptoms and Treatment of Frostbite

Symptoms include:

- Skin that becomes white, gray, or waxy yellow.
- Skin tingles and then becomes numb.
- Pain
- Blisters
- The area of frostbite swells and feels hard.

To treat frostbite, protect the area from refreezing. Apply warm (not hot) water to the affected area. Get medical attention immediately.



3.2.4 Hot Weather

- Heat exhaustion is characterized by pale, clammy skin; heavy sweating with nausea and possible vomiting; a fast, weak pulse; and possible fainting.
- Heat cramps can occur after an attack of heat exhaustion.
- Heat stroke is an immediate, life-threatening emergency that requires urgent medical attention. Its symptoms are headache, nausea, and visual problems. Body temperature can reach as high as 106° Fahrenheit. This is accompanied by hot, flushed, dry skin; slow, deep breathing; possible convulsions; and loss of consciousness.



3.3.0 Equipment Maintenance Safety

- Heavy equipment breakdowns can often be traced to improper maintenance or lack of regular maintenance.
- Only qualified, trained workers should be performing maintenance.



3.3.1 Pre-Maintenance Safety

- Wear appropriate PPE.
- If the work site does not have a designated service area, select a clean, level area out of the flow of traffic.
- Make sure there is enough room to work, as well as adequate light, ventilation, and overhead clearance.
- Use warning tags or signs as needed.



3.3.2 Engine Maintenance

- Do not work on a running engine unless necessary.
- If the engine must be running, work as a two-person team.
- Make sure you have proper ventilation from exhaust fumes.



3.3.3 Hydraulic System Maintenance

- Cycle hydraulic controls such as steering after shutdown to relieve trapped pressure in the lines before performing maintenance.
- Hot hydraulic fluid can cause severe burns. Wait for fluid to cool before disconnecting lines.
- Recharge hydraulic accumulators with dry nitrogen gas only.

Hydraulic fluid under pressure can penetrate skin or eyes and can cause injury, blindness, or death.



3.3.4 Cooling System Maintenance

- Before removing the radiator cap, stop the engine and let the system cool.
- Wear PPE.
- Slowly turn the radiator cap to the first position to allow any residual pressure to escape before removing the cap entirely.



3.3.5 Fueling

- Shut down equipment when filling the fuel tank.
- Always ground the fuel nozzle or container spout against the filler neck to avoid sparks.
- Never overfill the fuel tank.
- Clean up spilled fuel immediately.
- Keep sparks and open flames away from fuel.
- Do not smoke around fuel.
- Never cut or weld on fuel lines, tanks, or containers.
- Do not walk away when fueling equipment.



3.3.6 Battery Maintenance

- Keep arcs, sparks, and open flames away from batteries.
- Wear PPE to protect against battery fluid.
- Work on battery maintenance as a two-person team.
- Dispose of old batteries properly.



3.3.7 Tire Maintenance

- Properly inflated tires are important for equipment stability.
- Inspect tires and rims regularly for damage.



3.3.8 Air Conditioning System Maintenance

- Do not smoke or have open flames in the area when servicing the air conditioning system—refrigerant gases may be present.
- Refrigerant use is governed by the EPA.
- Only qualified technicians may work on air conditioning equipment.



3.3.9 Post-Repair Maintenance

- Complete all servicing and repairs before releasing any equipment.
- Make sure that all components are reinstalled and tested according to the manufacturer's recommendations.



Wrap Up – Trade Terms

Center of gravity: The point where all of an object's weight is evenly distributed.

Electrolyte: A substance that has the capacity to conduct electricity.

Pinch points: The area in which two moving equipment parts come together.



3.0.0 Section Review Question 1

1. If you are operating equipment that comes into contact with a power line you should _____.
 - a. stay in the cab until it is safe to exit unless the cab is on fire
 - b. immediately jump from the cab to the ground
 - c. jump down to the ground and run
 - d. try to remove the power line without leaving the cab



3.0.0 Section Review Question 1 Answer

1. If you are operating equipment that comes into contact with a power line you should _____.
 - a. **stay in the cab until it is safe to exit unless the cab is on fire**
 - b. immediately jump from the cab to the ground
 - c. jump down to the ground and run
 - d. try to remove the power line without leaving the cab



3.0.0 Section Review Question 2

2. Which of the methods for treating frostbite is recommended?
- a. Place the victim in a hot bath.
 - b. Apply heat directly to the affected area.
 - c. Give the victim whiskey or brandy.
 - d. Apply warm water to the affected area.



3.0.0 Section Review Question 2 Answer

2. Which of the methods for treating frostbite is recommended?
- a. Place the victim in a hot bath.
 - b. Apply heat directly to the affected area.
 - c. Give the victim whiskey or brandy.
 - d. Apply warm water to the affected area.**



3.0.0 Section Review Question 3

3. The reason to be careful about the refrigerant when working on a vehicle's air conditioning equipment is that _____.
- a. all refrigerant is poisonous
 - b. releasing refrigerants to the air is illegal
 - c. refrigerant is very expensive to replace
 - d. refrigerants are highly explosive



3.0.0 Section Review Question 3 Answer

3. The reason to be careful about the refrigerant when working on a vehicle's air conditioning equipment is that _____.
- a. all refrigerant is poisonous
 - b. releasing refrigerants to the air is illegal**
 - c. refrigerant is very expensive to replace
 - d. refrigerants are highly explosive



Next...

Section 4.0.0

Trenching and Excavation Safety



*Read Sections 4.0.0 through 4.3.3 and
complete the Section Review.*